SG No. 138 D_{4h}^{16} $P4_2/ncm$ [tetragonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `4a`, site symmetry: `2.22`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,-4,5,-6]`
3	$[-X, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,-10,15,-16]`
4	$[X, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11,-12,13,-14]`

Table 2: Wyckoff bond: `4b@4a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,-2,-7,8]`
2	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,-4,-5,6]`
3	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,-10,-15,16]`
4	$[-X, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11,-12,-13,14]`

Table 3: Wyckoff bond: `4c@4a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,2,-7,-8]`
2	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,4,-5,-6]`
3	$[0, 0, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,10,-15,-16]`
4	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11,12,-13,-14]`

Table 4: Wyckoff bond: `8d@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4,-5]`
5	$[-X, -X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,-16]`
6	$[X, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,-15]`
7	$[X, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11,-14]`
8	$[-X, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[12,-13]`

Table 5: Wyckoff bond: `8e@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[2,-8]`

continued ...

Table 5

No.	vector	center	mapping
3	$[X, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4,-6]`
5	$[-X, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,-15]`
6	$[X, -X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,-16]`
7	$[-X, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11,-13]`
8	$[X, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[12,-14]`

Table 6: Wyckoff bond: `8f@4a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[15,-16]`

Table 7: Wyckoff bond: `16g@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[16]`

* Wyckoff site: `4b`, site symmetry: `-4..`

Table 8: Wyckoff bond: `4a@4b`, bond symmetry: `-4..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,2,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,4,-9,-10]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,6,-15,-16]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7,8,-13,-14]`

Table 9: Wyckoff bond: `8b@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,-2]`
2	$[-Y,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,-4]`
3	$[-X,\ Y,\ 0]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,-6]`
4	$[Y,\ X,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7,-8]`
5	$[-X,\ -Y,\ 0]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,-10]`
6	$[Y,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[11,-12]`
7	$[X,\ -Y,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,-14]`
8	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[15,-16]`

Table 10: Wyckoff bond: `16c@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7]`
8	$[-Y,\ -X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[8]`
9	$[-X,\ -Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10]`
11	$[Y,\ -X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[11]`
12	$[-Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[12]`
13	$[X,\ -Y,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13]`
14	$[-X,\ Y,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[14]`
15	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[15]`
16	$[Y,\ X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `4c`, site symmetry: `..2/m`

Table 11: Wyckoff bond: `4a@4c`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-8,-9,16]`
2	$[-X,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,-7,-10,15]`

continued ...

Table 11

No.	vector	center	mapping
3	$[-X, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-6,-11,14]`
4	$[X, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,-5,-12,13]`

Table 12: Wyckoff bond: `4b@4c`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,8,-9,-16]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,7,-10,-15]`
3	$[X, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,6,-11,-14]`
4	$[-X, -X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,5,-12,-13]`

Table 13: Wyckoff bond: `8c@4c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-9]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[8,-16]`

* Wyckoff site: `4d`, site symmetry: `..2/m`

Table 14: Wyckoff bond: `4a@4d`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, Z]$	$[0, 0, 0]$	`[1,-8,-9,16]`
2	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-7,-10,15]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-6,-11,14]`
4	$[X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-5,-12,13]`

Table 15: Wyckoff bond: `4b@4d`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[0, 0, 0]$	`[1,8,-9,-16]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,7,-10,-15]`
3	$[X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,6,-11,-14]`
4	$[-X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,5,-12,-13]`

Table 16: Wyckoff bond: `8c@4d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-9]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$[0, 0, 0]$	`[8,-16]`

* Wyckoff site: `4e`, site symmetry: `2.mm`

Table 17: Wyckoff bond: `4a@4e`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,-2,-15,16]`
2	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,-4,-13,14]`
3	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[5,-6,-11,12]`
4	$[X, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[7,-8,-9,10]`

Table 18: Wyckoff bond: `4b@4e`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2,15,16]`
2	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,4,13,14]`
3	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[7,8,9,10]`

Table 19: Wyckoff bond: `8c@4e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,16]`
2	$[-X, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[2,15]`
3	$[-X, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,14]`
4	$[X, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[4,13]`
5	$[-X, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[5,12]`
6	$[X, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[6,11]`
7	$[X, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[7,10]`
8	$[-X, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[8,9]`

Table 20: Wyckoff bond: `8d@4e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,-2]`

continued ...

Table 20

No.	vector	center	mapping
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[3,-4]
3	$[-X, Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[5,-6]
4	$[Y, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[7,-8]
5	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[9,-10]
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[11,-12]
7	$[X, -Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[13,-14]
8	$[-Y, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[15,-16]

Table 21: Wyckoff bond: 16e@4e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[2]
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[3]
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[4]
5	$[-X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[5]
6	$[X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[6]
7	$[Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[7]
8	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[8]
9	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[9]
10	$[X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[10]
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[11]
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[12]
13	$[X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[13]
14	$[-X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[14]
15	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[15]
16	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[16]

* Wyckoff site: 8f, site symmetry: 2..

Table 22: Wyckoff bond: 8a@8f, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	[1,-2]
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[3,-4]
3	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[5,-6]
4	$[Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	[7,-8]
5	$[-X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	[9,-10]
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	[11,-12]
7	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[13,-14]
8	$[-Y, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	[15,-16]

Table 23: Wyckoff bond: `8b@8f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,2]`
2	$[0,\ 0,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[3,4]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[5,6]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[7,8]`
5	$[0,\ 0,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[9,10]`
6	$[0,\ 0,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[11,12]`
7	$[0,\ 0,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[13,14]`
8	$[0,\ 0,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[15,16]`

Table 24: Wyckoff bond: `16c@8f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[7]`
8	$[-Y,\ -X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[8]`
9	$[-X,\ -Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[10]`
11	$[Y,\ -X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[11]`
12	$[-Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[12]`
13	$[X,\ -Y,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[13]`
14	$[-X,\ Y,\ Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[14]`
15	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[15]`
16	$[Y,\ X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[16]`

* Wyckoff site: `8g`, site symmetry: `..2`

Table 25: Wyckoff bond: `8a@8g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ Z]$	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[1,-8]`
2	$[-X,\ -X,\ Z]$	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,-7]`
3	$[-X,\ X,\ Z]$	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[3,-6]`
4	$[X,\ -X,\ Z]$	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[4,-5]`
5	$[-X,\ -X,\ -Z]$	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[9,-16]`
6	$[X,\ X,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[10,-15]`
7	$[X,\ -X,\ -Z]$	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[11,-14]`
8	$[-X,\ X,\ -Z]$	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[12,-13]`

Table 26: Wyckoff bond: `8b@8g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[x, -x, \frac{1}{2}\right]$	`[1,8]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[2,7]`
3	$[X, X, 0]$	$\left[x+\frac{1}{2}, x, 0\right]$	`[3,6]`
4	$[-X, -X, 0]$	$\left[-x, \frac{1}{2}-x, 0\right]$	`[4,5]`
5	$[-X, X, 0]$	$\left[-x, x, \frac{1}{2}\right]$	`[9,16]`
6	$[X, -X, 0]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[10,15]`
7	$[-X, -X, 0]$	$\left[\frac{1}{2}-x, -x, 0\right]$	`[11,14]`
8	$[X, X, 0]$	$\left[x, x+\frac{1}{2}, 0\right]$	`[12,13]`

Table 27: Wyckoff bond: `16c@8g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[x+\frac{1}{2}, x, 0\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[-x, \frac{1}{2}-x, 0\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}-x, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, x, 0\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}-x, -x, 0\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[x, x+\frac{1}{2}, 0\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, x+\frac{1}{2}, 0\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, -x, 0\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[16]`

* Wyckoff site: `8h`, site symmetry: `..2`

Table 28: Wyckoff bond: `8a@8h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$[x, -x, 0]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, 0\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[4,-5]`
5	$[-X, -X, -Z]$	$[-x, x, 0]$	`[9,-16]`
6	$[X, X, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, 0\right]$	`[10,-15]`
7	$[X, -X, -Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{2}\right]$	`[11,-14]`
8	$[-X, X, -Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[12,-13]`

Table 29: Wyckoff bond: `8b@8h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, -x, 0]$	`[1,8]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, 0\right]$	`[2,7]`
3	$[X, X, 0]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}\right]$	`[3,6]`
4	$[-X, -X, 0]$	$\left[-x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[4,5]`
5	$[-X, X, 0]$	$[-x, x, 0]$	`[9,16]`
6	$[X, -X, 0]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, 0\right]$	`[10,15]`
7	$[-X, -X, 0]$	$\left[\frac{1}{2}-x, -x, \frac{1}{2}\right]$	`[11,14]`
8	$[X, X, 0]$	$\left[x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[12,13]`

Table 30: Wyckoff bond: `16c@8h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, -x, 0]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, 0\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, 0\right]$	`[7]`
8	$[-Y, -X, -Z]$	$[x, -x, 0]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, x, 0]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, 0\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{2}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, 0\right]$	`[15]`
16	$[Y, X, Z]$	$[-x, x, 0]$	`[16]`

* Wyckoff site: `8i`, site symmetry: `..m`

Table 31: Wyckoff bond: `8a@8i`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$[x, x, z]$	`[1,16]`
2	$[-X, -X, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	`[2,15]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}-x, x, z+\frac{1}{2}\right]$	`[3,14]`
4	$[X, -X, Z]$	$\left[x, \frac{1}{2}-x, z+\frac{1}{2}\right]$	`[4,13]`
5	$[-X, X, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[5,12]`
6	$[X, -X, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{2}-z\right]$	`[6,11]`
7	$[X, X, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[7,10]`
8	$[-X, -X, -Z]$	$[-x, -x, -z]$	`[8,9]`

Table 32: Wyckoff bond: `8b@8i`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, z]$	`[1,-16]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	`[2,-15]`
3	$[X, X, 0]$	$\left[\frac{1}{2}-x, x, z+\frac{1}{2}\right]$	`[3,-14]`
4	$[-X, -X, 0]$	$\left[x, \frac{1}{2}-x, z+\frac{1}{2}\right]$	`[4,-13]`
5	$[-X, -X, 0]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[5,-12]`
6	$[X, X, 0]$	$\left[x+\frac{1}{2}, -x, \frac{1}{2}-z\right]$	`[6,-11]`
7	$[-X, X, 0]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[7,-10]`
8	$[X, -X, 0]$	$[-x, -x, -z]$	`[8,-9]`

Table 33: Wyckoff bond: `16c@8i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}-x, x, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x, \frac{1}{2}-x, z+\frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{2}-z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$[-x, -x, -z]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -x, -z]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{2}-z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, \frac{1}{2}-x, z+\frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x, z+\frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	`[15]`
16	$[Y, X, Z]$	$[x, x, z]$	`[16]`

* Wyckoff site: `16j`, site symmetry: `1`

Table 34: Wyckoff bond: `16a@16j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}-y, x, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y, \frac{1}{2}-x, z+\frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, -y, \frac{1}{2}-z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[y+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$[-y, -x, -z]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, y+\frac{1}{2}, -z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[y+\frac{1}{2}, -x, \frac{1}{2}-z\right]$	`[11]`

continued ...

Table 34

No.	vector	center	mapping
12	$[-Y,\, X,\, -Z]$	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[12]
13	$[X,\, -Y,\, Z]$	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[13]
14	$[-X,\, Y,\, Z]$	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	[14]
15	$[-Y,\, -X,\, Z]$	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[15]
16	$[Y,\, X,\, Z]$	$[y,\, x,\, z]$	[16]